UNITED GROUP OF INSTITUTIONS
Affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow

SUMMER TRAINING PRESENTATION

Mental Health Prediction System

Predicting student wellbeing from social media use and lifestyle habits (web app: MindPulse AI)

PALAK SRIVASTAVA
Roll No. 2400100100310

Outline

01 **Introduction**
Why student wellbeing needs early screening

02 **Objectives**
What the system sets out to do

03 **Methodology Used**
Dataset, preprocessing and analysis

04 **Implementation**
Models, results and the MindPulse AI app

05 **Conclusion**
Outcome, limits and future scope

06 **References**
Sources and tools used

Introduction

Problem Statement

- Students handle study pressure, irregular sleep and long social media hours at the same time
- Self-reports and occasional counselling often spot strain late
- Without an early signal, students seek help only when problems are already serious

Proposed Solution

A machine learning system that estimates a student's Mental Health Score (about 1 to 10) from daily habits and personal details.
Along with the score it gives a wellbeing category, personalised tips and a counsellor alert for low scores.



+ 8 more inputs: age, gender, study hours, phone unlocks, platform and others

Objectives

1 Predict Mental Health Score

Estimate a student's score from social media use, sleep, study, activity and stress.

2 Analyse Influencing Habits

Find which habits are linked to better or poorer wellbeing.

3 Compare ML Algorithms

Train Linear Regression and Random Forest (default and tuned) on the same data.

4 Select the Best Model

Choose the top model using MAE, RMSE and R^2 on test data.

5 Build the MindPulse AI App

Serve predictions through a FastAPI backend with strict input checks.

6 Support Early Help-Seeking

Show a wellbeing category, tips and a counsellor alert with helplines.

Methodology Used

1. Problem Identification
2. Dataset Collection
3. Data Preprocessing
4. Exploratory Analysis
5. Feature Engineering
6. Model Training
7. Model Evaluation
8. Deployment (FastAPI)

Dataset

4,998 student records used

- Kaggle: Student Social Media and Mental Health Impact
- 13 columns: 12 inputs and 1 target
- Target: Mental Health Score (observed 3.6 to 9.4)
- Inputs: profile, social media use, sleep, study, activity, stress

Preprocessing

- Checked missing values and duplicates
- Study hours: log transform, then scaling
- Other numbers: standard scaling
- Stress level: ordered encoding (Low to Very High)
- Other categories: one-hot encoding; country kept to top 10 plus Other
- All steps packed in one scikit-learn pipeline

Exploratory Analysis

r = –0.82

daily social media hours vs score

- Sleep (+0.77) and study hours (+0.75) go with higher scores
- Median score falls from 8.0 (low stress) to 5.0 (very high stress)
- Shows association, not cause

Implementation

User Input
12 habit and profile details, validated

Processing
Country grouped; scaling and encoding applied

Random Forest
100 trees, trained regression model

Predicted Score
1 to 10, with a confidence level

Insights
Category, tips, counsellor alert, what-if

Models compared on the same held-out test data

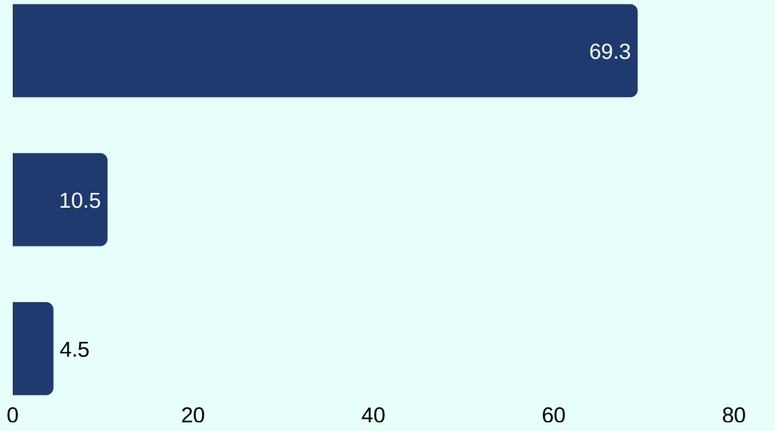
Model	Train R^2	Test R^2	MAE	RMSE
Linear Regression	0.724	0.740	0.536	0.678
Random Forest (default) ★	**0.981**	**0.878**	**0.347**	**0.464**
Random Forest (tuned)	0.955	0.865	0.369	0.487

Train R^2 0.981 vs test R^2 0.878 shows some overfitting; more validation is needed.

Best model: Random Forest (default)
Lowest error and highest R^2. Explains about 88% of score variation; average error about 0.35 points.

What drives the score

Top 3 of 12 features (importance %)



Screen time dominates; sleep is a distant second.

Conclusion

Achievements

- ✓ Collected and prepared the student dataset
- ✓ Completed preprocessing and EDA
- ✓ Trained and compared three ML models
- ✓ Selected Random Forest (R^2 0.878, MAE 0.347)
- ✓ Built MindPulse AI with FastAPI and input validation
- ✓ Added category, tips, counsellor alert and what-if tool

Outcome

Helps students and counsellors notice wellbeing strain early from everyday habits.

A screening and awareness tool only, not a clinical diagnosis.

Future Scope

- Larger, verified datasets
- XGBoost and LightGBM models
- Calibrated confidence level
- Campus counselling integration
- User accounts to track progress
- Android and iOS apps
- Automated tests, Docker, monitoring
- Advice checked by professionals

References

[1] Kaggle, “Student Social Media And Mental Health Impact” dataset. kaggle.com

[2] F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” J. Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[3] L. Breiman, “Random Forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[4] J. M. Twenge and W. K. Campbell, “Associations between screen time and lower psychological well-being among children and adolescents,” Preventive Medicine Reports, vol. 12, pp. 271–283, 2018.

[5] FastAPI Documentation. https://fastapi.tiangolo.com

[6] Pydantic Documentation. https://docs.pydantic.dev